

AMATH 403 HW 6

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Question 1: (6.2.4) The boundary value problem $\frac{d}{dx} \left(c(x) \frac{du}{dx} \right) = f(x)$, $u(0) = u(1) = 0$ models the displacement $u(x)$ of a nonuniform elastic bar with stiffness $c(x) = \frac{1}{1+x^2}$ for $0 \leq x \leq 1$

(a) Find the displacement when the bar is subjected to a constant external force, $f \equiv 1$

$$\begin{aligned} \frac{d}{dx} \left(c(x) \frac{du}{dx} \right) &= f(x) \\ \frac{d}{dx} \left(\frac{1}{1+x^2} \frac{du}{dx} \right) &= f(x) \\ \int \frac{d}{dx} \left(\frac{1}{1+x^2} \frac{du}{dx} \right) &= \int f(x) \\ f(x) &= 1 \\ \int \frac{d}{dx} \left(\frac{1}{1+x^2} \frac{du}{dx} \right) &= \int 1 \\ \frac{1}{1+x^2} \frac{du}{dx} &= x + C \\ \frac{du}{dx} &= (x + C)(1 + x^2) \\ \frac{du}{dx} &= x + x^3 + C + Cx^2 \\ u(x) &= \frac{x^2}{2} + \frac{x^4}{4} + Cx + C\frac{x^3}{3} + D \end{aligned}$$

Initial Conditions

$$\begin{aligned} u(0) &= \frac{0^2}{2} + \frac{0^4}{4} + C(0) + C\frac{0^3}{3} + D = 0 \\ D &= 0 \end{aligned}$$

$$\begin{aligned} u(1) &= \frac{1}{2} + \frac{1}{4} + C + C\frac{1}{3} = 0 \\ \frac{3}{4} + \frac{4}{3}C &= 0 \\ \frac{4}{3}C &= \frac{-3}{4} \\ C &= \frac{-9}{16} \end{aligned}$$

Substitute C and D

$$u(x) = \frac{x^2}{2} + \frac{x^4}{4} - \frac{9}{16}x - \frac{3}{16}x^2$$

(b) Find the Green's function for the boundary value problem.

$$\begin{aligned}\frac{d}{dx} \left(\frac{1}{1+x^2} \frac{dG}{dx} \right) &= \delta(x-\eta) \\ G(0, \eta) &= G(1, \eta) = 0\end{aligned}$$

$$\frac{1}{1+x^2} \frac{dG}{dx} = A \implies \frac{dG_1}{dx} = A(1+x^2)$$

$$G_1(x) = A\left(x + \frac{x^3}{3}\right) + C, \quad G_1(0) = 0$$

$$C = 0$$

$$G_1(x) = A\left(x + \frac{x^3}{3}\right)$$

$$G_2(x) = B\left(x + \frac{x^3}{3}\right) + D$$

$$G_2(1) = 0$$

$$B\left(1 + \frac{1}{3}\right) + D = 0 \implies \frac{4}{3}B + D = 0$$

$$D = -\frac{4}{3}B$$

$$G_2(x) = B\left(x + \frac{x^3}{3} - \frac{4}{3}\right)$$

$$G_1(\eta) = G_2(\eta)$$

$$A\left(\eta + \frac{\eta^3}{3}\right) = B\left(\eta + \frac{\eta^3}{3} - \frac{4}{3}\right)$$

Jump Condition

$$\int_{\eta-\epsilon}^{\eta+\epsilon} \frac{d}{dx} \left(\frac{1}{1+x^2} \frac{dG}{dx} \right) dx = \int_{\eta-\epsilon}^{\eta+\epsilon} \delta(x-\eta) dx$$

$$\frac{1}{1+\eta^2} G_2'(\eta) - \frac{1}{1+\eta^2} G_1'(\eta) = 1$$

$$G_2'(\eta) - G_1'(\eta) = 1 + \eta^2$$

$$B - A = 1$$

$$B = A + 1$$

$$\begin{aligned}
A\left(\eta + \frac{\eta^3}{3}\right) &= (A+1)\left(\eta + \frac{\eta^3}{3} - \frac{4}{3}\right) \\
A\left(\eta + \frac{\eta^3}{3}\right) &= A\left(\eta + \frac{\eta^3}{3} - \frac{4}{3}\right) + \left(\eta + \frac{\eta^3}{3} - \frac{4}{3}\right) \\
&= A\left(\eta + \frac{\eta^3}{3}\right) - \frac{4}{3}A + \eta + \frac{\eta^3}{3} - \frac{4}{3} \\
0 &= -\frac{4}{3}A + \eta + \frac{\eta^3}{3} - \frac{4}{3} \\
\frac{4}{3} &= \eta + \frac{\eta^3}{3} - \frac{4}{3} \\
\frac{4}{3}A &= \eta + \frac{\eta^3}{3} - \frac{4}{3} \\
A &= \frac{3}{4}\eta + \frac{1}{4}\eta^3 - 1
\end{aligned}$$

$$B + \left(\frac{3}{4}\eta + \frac{1}{4}\eta^3 - 1\right) + 1 \implies B = \frac{3}{4}\eta + \frac{1}{4}\eta^3$$

Final $G(x, \eta)$

$$G(x, \eta) = \begin{cases} \left(\frac{3}{4}\eta + \frac{1}{4}\eta^3 - 1\right) \left(x + \frac{x^3}{3}\right) & \text{for } 0 \leq x \leq \eta \\ \left(\frac{3}{4}\eta + \frac{1}{4}\eta^3\right) \left(x + \frac{x^3}{3} - \frac{4}{3}\right) & \text{for } \eta < x \leq 1 \end{cases}$$

Question 2: (7.2.7) If $f(x)$ has Fourier transform $\hat{f}(k)$, which function has Fourier transform $\frac{\hat{f}(k)}{k}$

We want to find $\hat{g}(k) = \frac{\hat{f}(k)}{k}$

We can use the derivative property here:

$$\mathcal{F}(g') = ik\hat{g}(k)$$

$$\hat{g}(k) = \frac{\hat{f}(k)}{k}$$

Multiply by ik

$$ik\hat{g}(k) = i\hat{f}(k)$$

Substitute $\mathcal{F}(g') = ik\hat{g}(k)$

$$\mathcal{F}(g') = \mathcal{F}(if(x))$$

The Fourier transformations are equal on both sides so the functions must be equal

$$g'(x) = if(x)$$

We can integrate both sides

$$\int g'(x) dx = i \int f(x) dx$$

$$g(x) = i \int f(x) dx$$

Our function is $g(x) = i \int f(x) dx$ where $\hat{g}(k) = \frac{\hat{f}(k)}{k}$.

Question 3: (7.3.8) Use the Fourier transform to find an integral formula for a bounded solution to the Airy differential equation $\frac{d^2u}{dx^2} = xu$

$$\begin{aligned}\frac{d^2u}{dx^2} &= xu \\ \mathcal{F}(u'') &= \mathcal{F}(xu)\end{aligned}$$

Using the derivative property

$$\mathcal{F}(u'') = (ik)^2 \hat{u}(k) = -k^2 \hat{u}(k)$$

$$\begin{aligned}\hat{u}(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u(x) e^{-ikx} dx \\ \frac{d\hat{u}}{dk} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (-ix) u(x) e^{-ikx} dx\end{aligned}$$

Divide both sides by $-i$

$$\frac{id\hat{u}}{dk} = \mathcal{F}(xu)$$

Substitute $\mathcal{F}(xu) = -k^2 \hat{u}(k)$

$$-k^2 \hat{u}(k) = \frac{id\hat{u}}{dk}$$

Solve ODE

$$-k^2 \hat{u}(k) = \frac{id\hat{u}}{dk}$$

$$\frac{d\hat{u}}{\hat{u}} = ik^2 dk$$

$$\ln(\hat{u}) = i\frac{k^3}{3} + C$$

$$\hat{u}(k) = Ae^{i\frac{k^3}{3}}$$

$$u(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} Ae^{i\frac{k^3}{3}} e^{ikx} dk$$

$$u(x) = \frac{A}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\frac{k^3}{3}} e^{ikx} dk$$